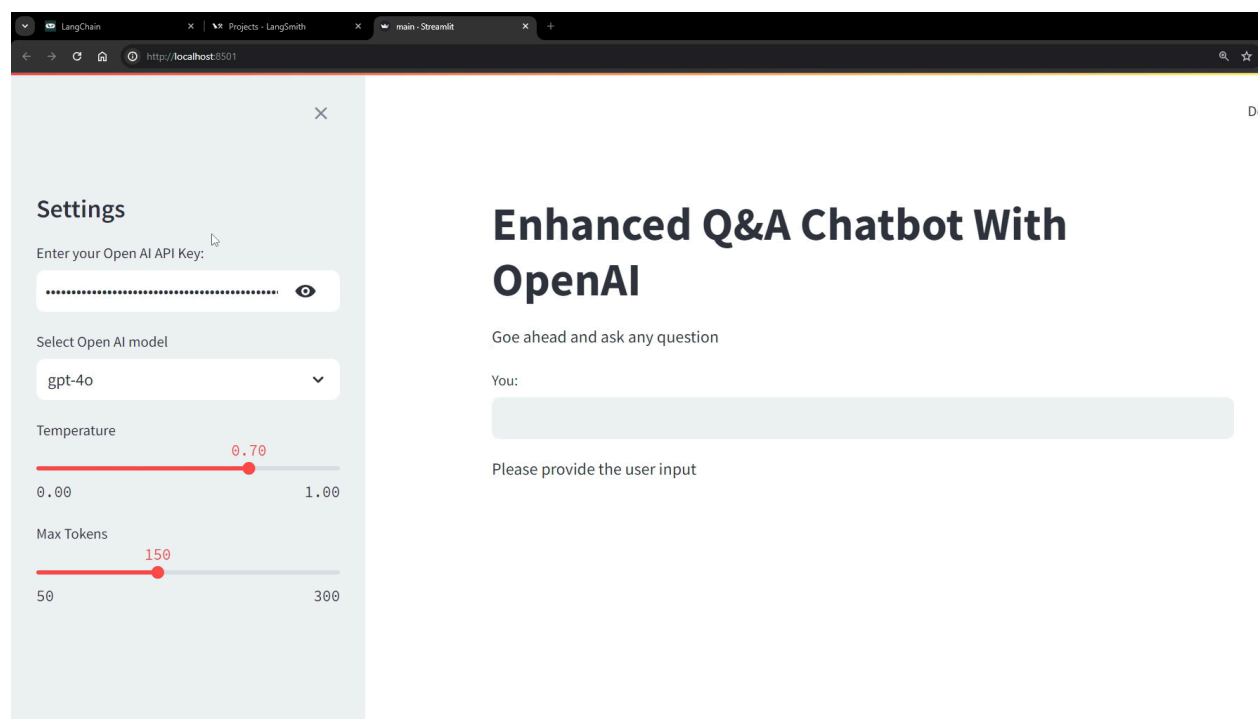
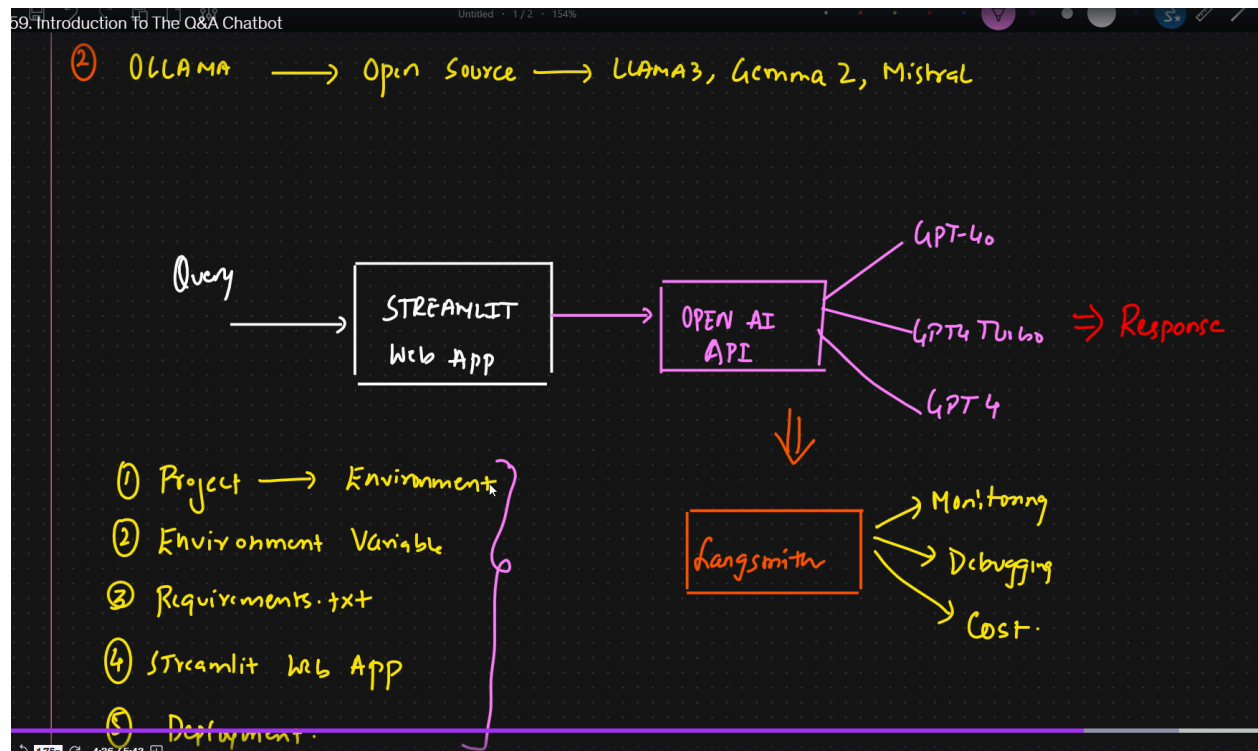


Here we build a project with the following tools

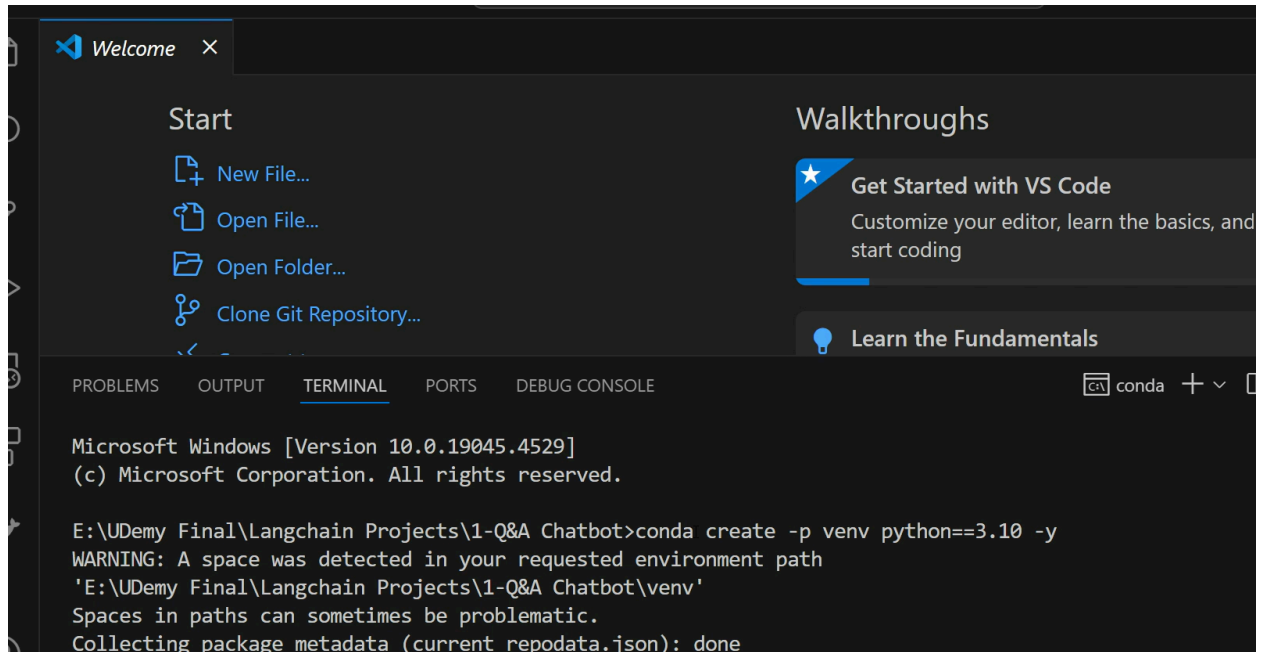
Q&A CHATBOT [End to End Gen AI Project]

① Open AI LLM Models → GPT-4o, GPT-4 Turbo

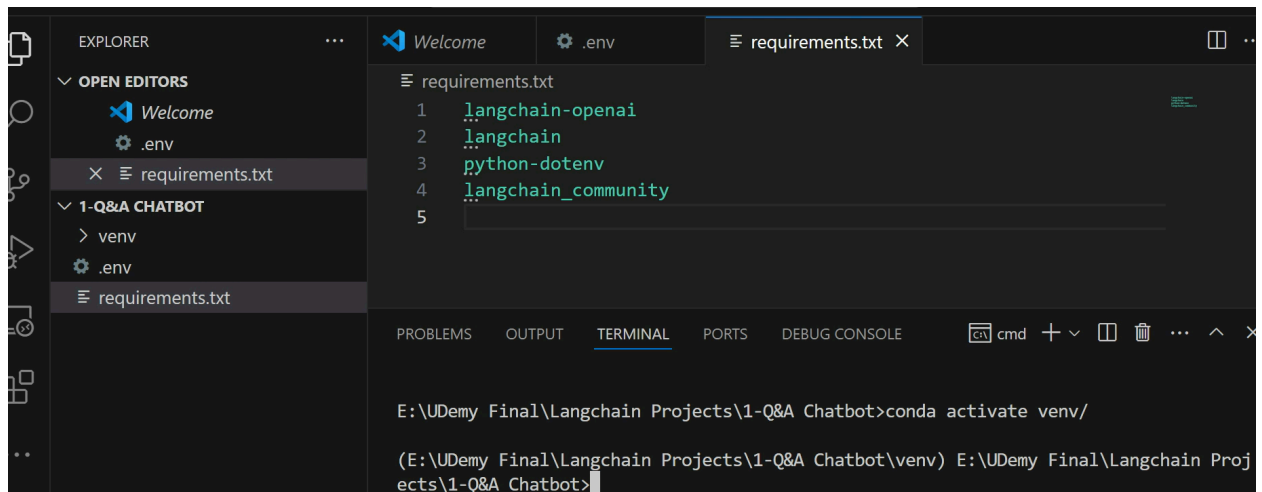
② OLLAMA → Open Source → LLAMA3, Gemma 2, Mistral

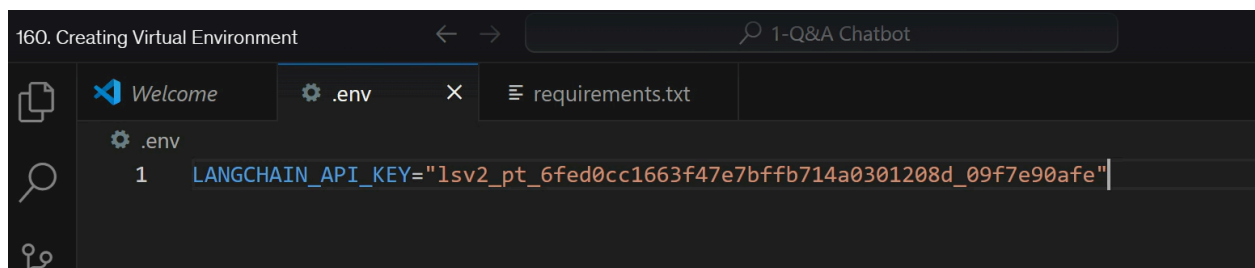
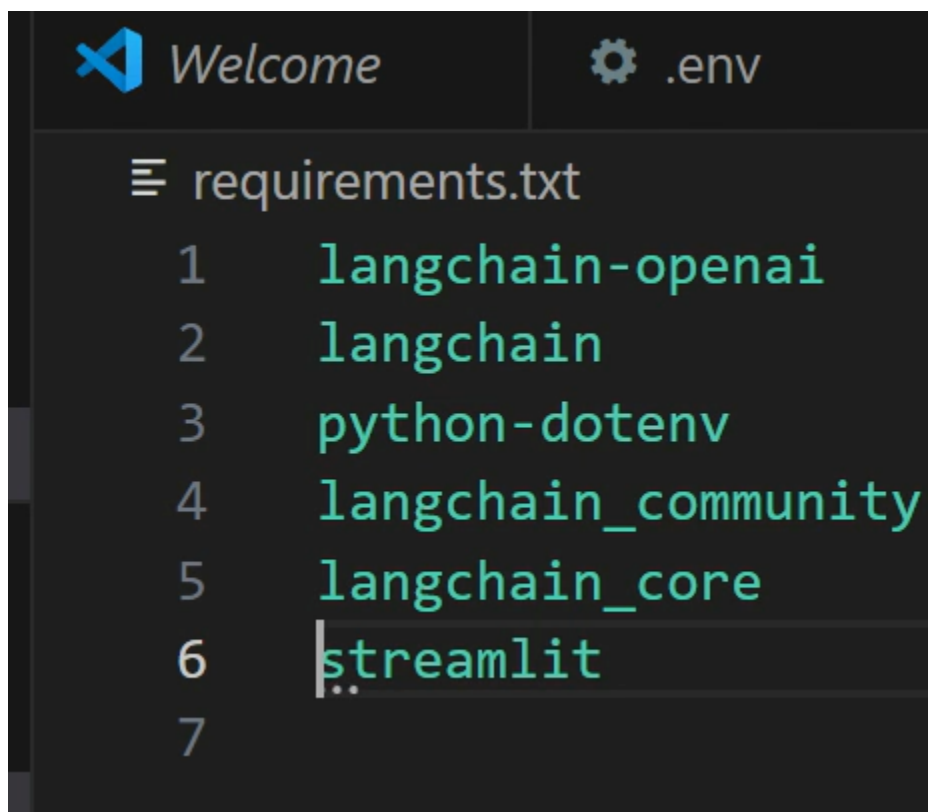
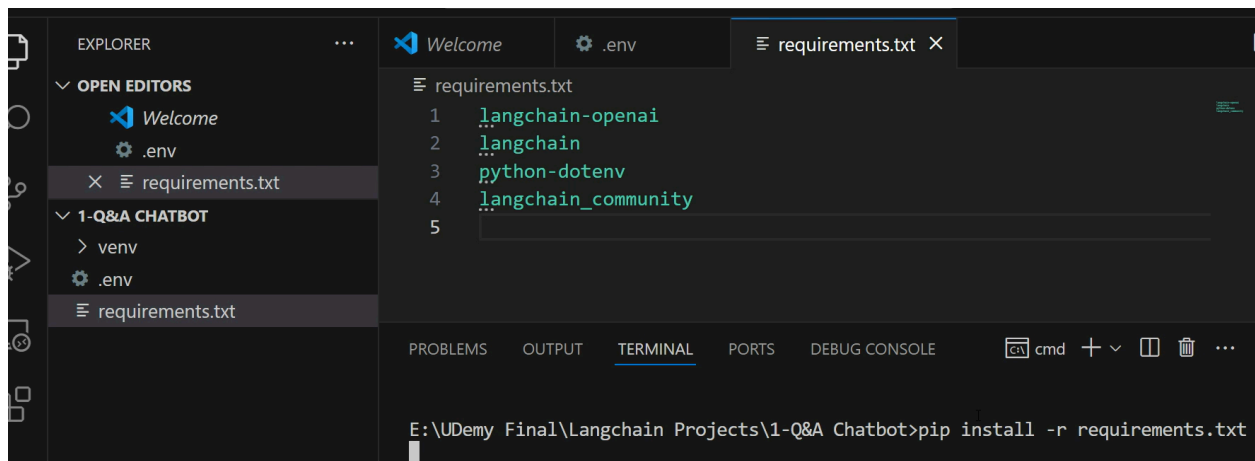


lets create a project



Activate Venv and install the requirements





Difference between openAI and Ollama

Feature	OpenAI	Ollama
Where it runs	Cloud (internet required)	Local (your machine)
Setup	No setup, just use	Requires install + setup
Cost	Pay-per-use / subscription	Free (but needs good hardware)
Performance	Very high (server-grade GPUs)	Depends on your PC
Privacy	Data sent to servers	Data stays on your device
Ease of use	Beginner-friendly	More technical
Models	Proprietary (GPT-4, etc.)	Open-source (Llama, Mistral, etc.)

Where as langchain is a frame that can interact with openai, huggingface, opensource LLM models, etc

here temperature is the creativity of the model, 0 is low creativity, and 1 for high creativity.

```

1  import streamlit as st
2  import openai
3  from langchain_openai import ChatOpenAI
4  from langchain_core.output_parsers import StrOutputParser
5  from langchain_core.prompts import ChatPromptTemplate
6  import os
7
8  import os
9  from dotenv import load_dotenv
10 load_dotenv()

```

Chain is something how interaction will happen.

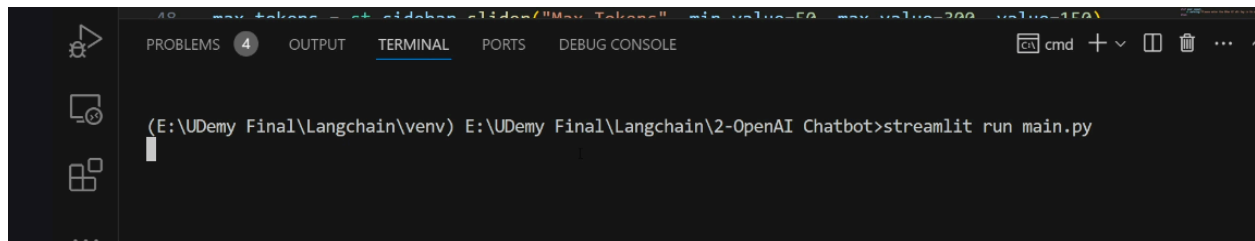
```

15 os.environ["LANGCHAIN_TRACING_V2"] = True
16 os.environ["LANGCHAIN_PROJECT"] = "Q&A Chatbot With OPENAI"
17
18 ## Prompt Template
19 prompt=ChatPromptTemplate.from_messages(
20     [
21         ("system", "You are a helpful assistant. Please response to the user queries"),
22         ("user", "Question:{question}")
23     ]
24 )
25
26 def generate_response(question, api_key, llm, temperature, max_tokens):
27     openai.api_key=api_key
28     llm=ChatOpenAI(model=llm)
29     output_parser=StrOutputParser()
30     chain=prompt|llm|output_parser
31     answer=chain.invoke({'question':question})
32     return answer

```

look code for the rest of the implementation.

to run the app



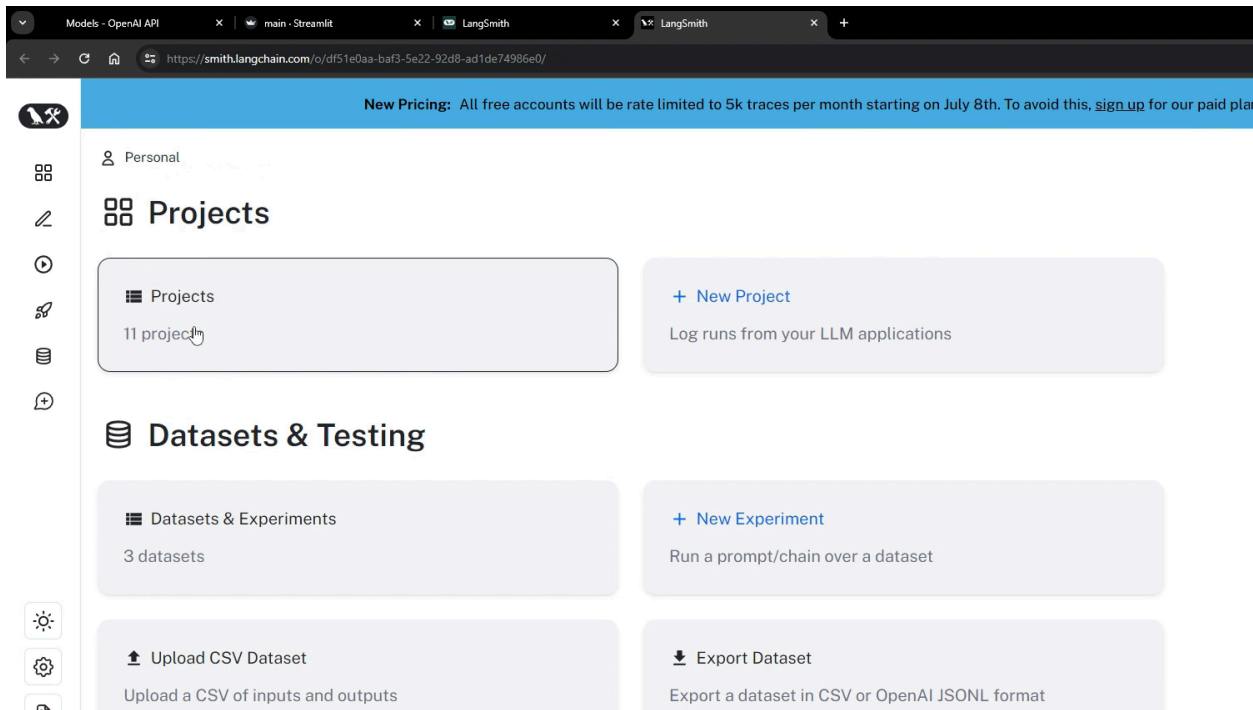
The screenshot shows a VS Code interface with a terminal window open. The terminal shows the command `streamlit run main.py` being executed in a virtual environment. The file explorer on the left shows a project structure with a `main.py` file.

```

(E:\UDemy Final\Langchain\venv) E:\UDemy Final\Langchain\2-OpenAI Chatbot>streamlit run main.py

```

You can check the project status



The screenshot shows the 'Projects' table in the LangSmith dashboard. The table has columns for Name, Feedback (7D), Run Count (7D), Error Rate (7D), % Streaming (7D), Total Tokens (7D), Total Cost (7D), and P50 Latency (7D). There are two projects listed: 'Simple Q&A Chatbot With OPENAI' and 'Simple Q&A Chatbot With OLLAMA'.

Name	Feedback (7D)	Run Count (7D)	Error Rate (7D)	% Streaming (7D)	Total Tokens (7D)	Total Cost (7D)	P50 Latency (7D)
Simple Q&A Chatbot With OPENAI		8	0%	0%	4,388	\$0.06313	9.10s
Simple Q&A Chatbot With OLLAMA		2	0%	100%	346		4.65s

The screenshot shows the 'Simple Q&A Chatbot With OPENAI' project page in the LangSmith dashboard. The 'Runs' tab is selected, showing a list of runs with columns for Name, Input, Start Time, Latency, Dataset, Annotation Queue, and Tokens. There are four runs listed: 'RunnableSequence', 'ChatPromptTemplate', 'ChatOpenAI', and 'StrOutputParser'.

Name	Input	Start Time	Latency	Dataset	Annotation Queue	Tokens
RunnableSequence	what iss machine learn...	7/2/2024, 3:05:18 PM	8.01s			406
ChatPromptTemplate	what iss machine learn...	7/2/2024, 3:05:18 PM	0.01s			0
ChatOpenAI	human: Question:what ...	7/2/2024, 3:05:18 PM	7.93s			406
StrOutputParser	ai: Machine learning is ...	7/2/2024, 3:05:26 PM	0.01s			0

you can click it and see the details

Simple Q&A Chatbot With Ollama

Runs Threads Monitor Setup

1 filter Last 7 days Root Runs LLM Calls All Runs

Name	Input
RunnableSequence	what is machine learning
ChatPromptTemplate	what is machine learning
ChatOpenAI	human: Question: what is machine learning
StrOutputParser	ai: Machine learning is applied in various domains such as...
RunnableSequence	please explain more
RunnableSequence	Hi
RunnableSequence	Explain about machine learning
RunnableSequence	Explain about machine learning

TRACE

RunnableSequence 8.01s 406

ChatOpenAI gpt-4o 7.93s

Some runs have been hidden. Show 2 hidden runs

ChatOpenAI

Run Feedback Metadata

HUMAN

Question: what is machine learning

Output

AI

Machine learning is applied in various domains such as...

6. **Model Evaluation**: Techniques to assess how well a machine learning model performs, such as cross-validation, confusion matrices, and various performance metrics like accuracy, precision, recall, and F1 score.

Implementing same thing with OLLAMA

```
1 import streamlit as st
2 import openai
3 from langchain_openai import ChatOpenAI
4 from langchain_core.output_parsers import StrOutputParser
5 from langchain_core.prompts import ChatPromptTemplate
6 from langchain_community.llms import Ollama
7 import os
8
9 import os
10 from dotenv import load_dotenv
11 load_dotenv()
12
13 ## Langsmith Tracking
14 os.environ["LANGCHAIN_API_KEY"] = os.getenv("LANGCHAIN_API_KEY")
15 os.environ["LANGCHAIN_TRACING_V2"] = "true"
16 os.environ["LANGCHAIN_PROJECT"] = "Simple Q&A Chatbot With Ollama"
17
18 ## Prompt Template
19 prompt = ChatPromptTemplate.from_messages(
20     [
21         ("system", "You are a helpful assistant. Please respond to the user's question."),
22         ("user", "Question: {question}")
23     ]
24 )
```

```
def generate_response(question, api_key, engine, temperature, max_tokens):
    llm = Ollama(model=engine)
    output_parser = StrOutputParser()
    chain = prompt | llm | output_parser
    answer = chain.invoke({'question': question})
    return answer
```


to check the models running in your local, run the following command. Which ever the libraries that you want to use with Ollama, first download those libraries to your local.

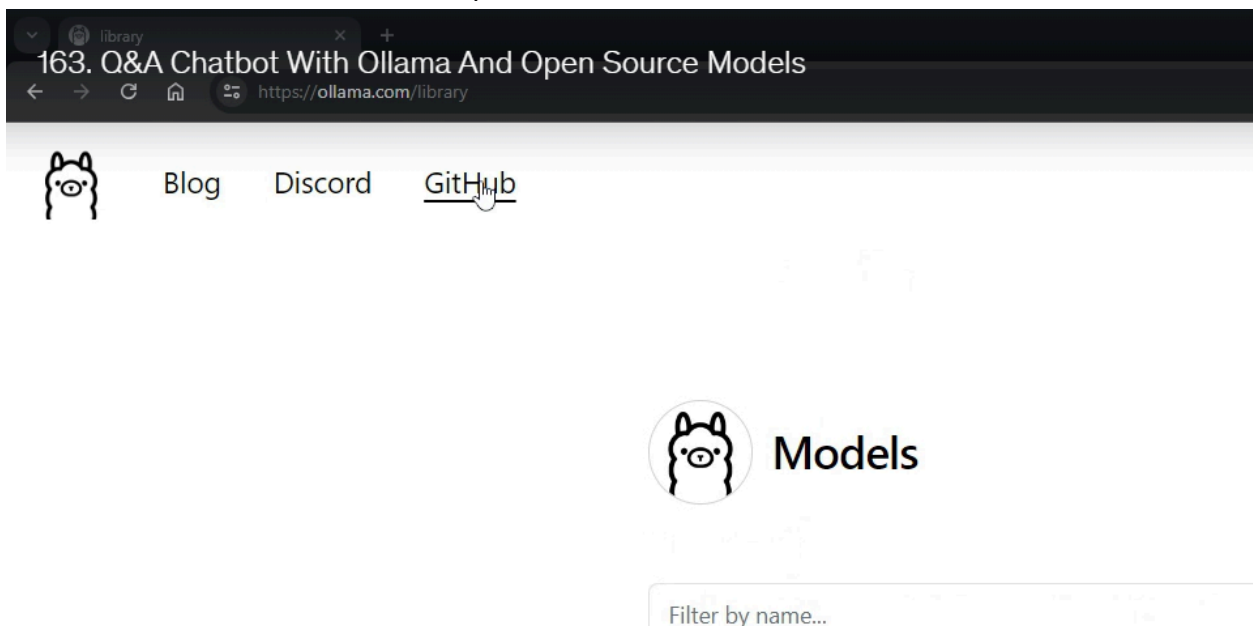
```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

C:\Users\win10>ollama run gemma2
pulling manifest
pulling 0da0a4d67fe4... 6% [ ]
```

```
C:\Users\win10>ollama phi3
Error: unknown command "phi3" for "ollama"

C:\Users\win10>ollama run phi3
pulling manifest [ ]
```

to check what all the models ollama provides to site and check



In the github you will get all the instructions.

163. Q&A Chatbot With Ollama And Open Source Models

https://github.com/ollama/ollama

README MIT license

Here are some example models that can be downloaded:

Model	Parameters	Size	Download
Llama 3	8B	4.7GB	<code>ollama run llama3</code>
Llama 3	70B	40GB	<code>ollama run llama3:70b</code>
Phi 3 Mini	3.8B	2.3GB	<code>ollama run phi3</code>
Phi 3 Medium	14B	7.9GB	<code>ollama run phi3:medium</code>
Gemma 2	9B	5.5GB	<code>ollama run gemma2</code>
Gemma 2	27B	16GB	<code>ollama run gemma2:27b</code>
Mistral	7B	4.1GB	<code>ollama run mistral</code>
Moondream 2	1.4B	829MB	<code>ollama run moondream</code>
Neural Chat	7B	4.1GB	<code>ollama run neural-chat</code>
Starling	7B	4.1GB	<code>ollama run starling-1m</code>
Code Llama	7B	3.8GB	<code>ollama run codellama</code>
Llama 2 Uncensored	7B	3.8GB	<code>ollama run llama2-uncensored</code>
LLaVA	7B	4.5GB	<code>ollama run llava</code>
Solar	10.7B	6.1GB	<code>ollama run solar</code>

Check the code, here we don't need api key, in the code. That is the one modified from previous one.

163. Q&A Chatbot With Ollama And Open Source Models

http://localhost:8501

Select Open AI model

mistral

Temperature

0.00 0.70 1.00

Max Tokens

50 150 300

Go ahead and ask any question

You:

Hello please talk about generative AI

Hello! I'd be happy to help answer your question about Generative AI

Generative AI is a subfield of artificial intelligence that focuses on creating new data based on those learned patterns. This can include generating text, images, or even code.

There are several types of generative models such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Generative Stochastic Neural Networks (GSNNs).

[https://smith.langchain.com/c/df51e0aa-baf3-5e22-92d8-ad1de74986e0/projects?paginationState=%7B%22pageIndex%3A0%2C%22pageSize%3A10%7D](#)

Personal > Projects

Projects

Columns

Name	Feedback (7D)	Run Count (7D)	Error Rate (7D)	% Streamed
Simple Q&A Chatbot With OLLAMA		2	0%	100%
Simple Q&A Chatbot With OPENAI		5	0%	0%
GenAIAPPWithOPENAI		0	0%	0%
Langchain series		0	0%	0%

163. Q&A Chatbot With Ollama And Open Source Models

New Pricing: All free accounts will be rate limited to 5k traces per month starting on July 8th. To avoid this, [sign up](#) for our paid plans

Personal > Projects > Simple Q&A Chatbot With OLL...

BETA USER

Simple Q&A Chatbot With OLLAMA

ID Data Retention 400d Add Rule Edit

Runs Threads Monitor Setup

1 filter Last 7 days Root Runs LLM Calls All Runs Columns

	>	✓ Name	Input	Start Time	Latency	Dataset	Annotation Queue	Tokens
✓ RunnableSequence		Hello please talk about...	7/2/2024, 11:49:09 A...	6.65s			301	
✓ RunnableSequence		Hi	7/2/2024, 11:48:53 ...	2.64s			45	

Stats

Last 7 d

RUN COU

2

TOTAL TO

346

MEDIAN T

173

ERROR R /

0%

% STREA I

100%

LATENCY

P50: 4.6S